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ORIGINAL

Combined use of a multiplex PCR and serum procalcitonin to reduce antibiotic exposure in critically ill patients with community-acquired pneumonia: the MULTI-CAP randomized controlled trial

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Abstract

Purpose: Multiplex polymerase chain reaction (mPCR) testing has the potential to rapidly and accurately identify causative microorganisms in patients with community-acquired pneumonia (CAP). Its use in a management strategy, along with biomarkers, may reduce antibiotic exposure and improve clinical outcomes.

Methods: The MULTI-CAP trial was a multicenter ($n=20$), parallel-group, superiority, open-label, randomized trial. Subjects were non-immunocompromised adult patients (≥ 18 years) admitted to the intensive care unit (ICU) for CAP and randomly assigned in a 1:1 ratio. In the intervention group, the microbiological diagnosis combined a broad-spectrum respiratory mPCR and conventional microbiological investigations. An algorithm for early discontinuation or de-escalation of antibiotics was applied, based on mPCR results and serum procalcitonin. In the control group, only conventional microbiological investigations were performed. In both groups, antibiotic discontinuation was considered on Day 3 and day after day until Day 7, based on procalcitonin values and kinetics. The primary endpoint was defined as the number of days alive without any antibiotic from the time of enrollment to Day 28.

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The MULTI-CAP Collaborative Trial Group Investigators are listed in the Acknowledgments.

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Results: From October 4, 2018, to March 3, 2022, 406 patients were randomized, and 385 were evaluable in the intention-to-treat analysis. The median number of days alive without antibiotics on Day 28 was 19.0 (0.0; 24.0) days in the intervention group and 19.0 (7.0; 22.0) days in the control group (difference, 0.0 (95% CI, − 4.0 to 4.0). However, the antibiotic cumulative duration on day 28 was 3 days shorter (95% CI, − 5.1 to − 0.9) in the intervention group. Serious adverse events did not differ between groups.

Conclusion: In ICU patients with CAP, a management strategy combining a mPCR and serum procalcitonin failed to reduce antibiotic exposure or improve outcomes on Day 28, compared to usual care.

Trial registration number: NCT03452826 (March 2018), EudraCT 2017-A01615-48.

Keywords: Community-acquired pneumonia, Critically ill, Intensive care medicine, Molecular diagnosis

Introduction

Community-acquired pneumonia (CAP) is one of the leading causes of intensive care unit (ICU) admission and in-hospital mortality. It is the main reason for antibiotic therapy in severe community-acquired infections. Inadequate or overly broad antibiotic therapy is associated with poor outcomes in sepsis and may promote bacterial resistance [1, 2]. Strategies incorporating adequate and narrow-spectrum antimicrobial therapy are needed. In recent decades, approaches aimed at personalizing the duration of antimicrobial therapy, guided by clinical cure in hospitalized patients [3] or the values and kinetics of the biomarker procalcitonin in ICU patients [4, 5], have shown promising results referenced in the recent European guidelines [6]. Another promising area for research is the improvement of microbiological diagnosis using multi-target nucleic acid amplification tests. Recently, multiplex polymerase chain reaction (mPCR) tests with enlarged respiratory panels have been developed and commercialized. These tests can accurately detect a large panel of respiratory microorganisms, including bacteria and viruses, as well as some bacterial resistance genes, in a short turnaround time. The routine use of this method in CAP patients, in addition to conventional tests, could allow rapid and accurate identification of the responsible microorganisms to guide therapy. However, clinical evidence of a beneficial impact on CAP is lacking. First, mPCR-related bacterial detection may facilitate antibiotic de-escalation, from empirical to targeted therapy, or from broad-spectrum to narrow-spectrum antibiotics [7–9]. Second, mPCR-related bacterial detection may increase the proportion of patients receiving appropriate therapy [10]. Third, mPCR-related "pure" viral detection in conjunction with a low procalcitonin level may encourage clinicians to stop antibiotics early in some patients [11].

Therefore, we developed an innovative strategy for the diagnosis and therapeutic management of CAP that combines a broad-spectrum respiratory mPCR and serum procalcitonin measurements, using an algorithm to guide early antibiotics discontinuation or de-escalation based on the outcomes. We conducted a multicenter, superiority, open-label, randomized trial to assess the efficacy and safety of this mPCR-procalcitonin algorithm approach, as compared with standard of care to help clinicians tailor antibiotics in the early management of critically ill patients with CAP. Our objective was to determine whether the combined use of respiratory molecular testing and biomarker values and kinetics would result in decreased antibiotic exposure and improved clinical outcomes, as reflected by an increased number of days alive without antibiotics from enrollment to Day 28.

Methods

Study design

The study protocol of the MULTI-CAP study has been previously published [12]. The protocol and statistical analysis plan are presented in Supplement 2 and Supplement 3. The MULTI-CAP trial is a national multicenter, parallel-group, open-label, superiority randomized controlled trial in critically ill patients with CAP. It compares the efficacy and safety of a diagnostic and therapeutic management strategy combining a broad-spectrum respiratory mPCR and an algorithm for early antibiotic discontinuation or de-escalation (*Experimental Strategy*) with the efficacy and safety of standard of care (*Usual Strategy*).

Ethical issues

The MULTI-CAP trial was conducted in accordance with the Declaration of Helsinki. This trial was approved by

the Committee for the Protection of Individuals (CPP Ile de France V) and the French National Agency of Medicine and Health Products Safety (ANSM), and registered in Clinical Trials (NCT03452826).

Participants

Participants were patients aged at least 18 years admitted to one of the 20 participating French ICUs. CAP was considered severe when ICU admission was decided by the attending physicians. The main exclusion criteria were immunocompromised status and a history of respiratory colonization with *Pseudomonas aeruginosa* (for details regarding enrollment and exclusion criteria, see Supplement 1).

Randomization

After obtaining written informed consent from the patient or next of kin, eligible patients were randomized using a web-based system and assigned either to the Usual strategy (control group) or to the Experimental strategy (intervention group) in a 1:1 ratio. Balanced-block randomization was computer-generated by an independent statistician and stratified by participating site.

Usual strategy

The usual strategy was based on the standard of care for CAP, in accordance with international guidelines [6, 13]. Conventional microbiological investigations, including urine antigen assays, blood cultures, and respiratory tract samples (for details regarding microbiological work-up, see Supplement 1) were at the attending physician's discretion and performed as soon as possible after ICU admission. Influenza monoplex PCR was encouraged during the flu season. Serum concentrations of procalcitonin (ThermoFisher Scientific®) were measured at randomization (Hour 0), twice on Day 1 (Hour 12 and Hour 24), and then daily from Day 3 to Day 7. Clinicians were encouraged to perform a chest X-ray at enrollment, but could opt for a chest CT-scan.

Patients may have received antibiotics before ICU referral. In other cases, antibiotics were initiated as soon as possible after ICU admission. Written guidelines for empirical and secondary antibiotics was available for clinicians (see Supplement 2 and Table S1, Supplement 1). The final decision for the choice of antimicrobials (drugs and dose regimen) was at the attending physician's discretion. Antibiotic discontinuation was considered on Day 3 and day after day until Day 7, based on procalcitonin values and kinetics, as previously described [14] (Figure S1, Supplement 1). The recommended maximum duration of antibiotics was 7 days, unless otherwise indicated. When microorganisms were identified, pathogen-targeted therapy was encouraged.

Experimental strategy

In the intervention group, a respiratory broad panel mPCR (BioFire® Film Array® Pneumonia Plus Panel) (see Supplement 1 for details) was performed on a lower respiratory tract sample collected within 12 h of enrollment, in addition to conventional microbiological investigations. Before the end of Day 1, clinicians had to consider all the available microbiological results (mPCR FA-PPP, urine antigen assays, blood cultures, and Gram stain examination of respiratory tract samples) and procalcitonin values. They subsequently had to apply an algorithm of early antibiotic discontinuation or de-escalation (Figure S2, Supplement 1). Discontinuation was recommended in the absence of bacterial documentation and a procalcitonin concentration < 1 ng/mL. Discontinuation was strongly recommended if concurrent viral documentation was obtained. Otherwise, antibiotics continuation was encouraged, with narrowing the spectrum as much as possible. Specifically, if no atypical bacteria were detected, discontinuation of macrolide (or fluoroquinolone) was strongly recommended. The adherence to the algorithm was collected prospectively from investigators (for details, see Supplement 1). On Day 3 and daily thereafter until Day 7, antibiotic discontinuation had to be considered, based on procalcitonin values and kinetics, as in the control arm.

Endpoints

The primary outcome was the number of antibiotic-free days at Day 28, defined as the number of days alive without antibiotics (including antibiotics administered for any reason, except prophylactic antibiotics), either parenteral or enteral, from enrollment to Day 28.

Secondary outcomes were mortality on Day 28 and Day 90, death or persistent organ failure(s) on Day 28, antibiotic exposure on Day 28, antibiotic cumulative duration on Day 28, incidence rates of colonization/infection with multidrug-resistant bacteria or *Clostridioides difficile* on Day 28, ICU and hospital length of stay, quality of life on Day 90, total cost of care and decremental cost-effectiveness ratio (Table S2, Supplement 1). Antibiotic exposure was measured as the number of days with at least one antibiotic regardless of route and dose regimen from enrollment to Day 28. Cumulative duration of antibiotics, as a measure of total antibiotic consumption, was calculated as the sum of the duration of each antibiotic from enrollment to Day 28 [15, 16].

Follow-up

Patients were followed for 90 days. If the patient was discharged before Day 28, vital status and post-discharge antibiotic exposure were ascertained through telephone communication with the patient or their next of kin. Similarly, information regarding hospital readmission was collected via telephone on Day 90.

Sample size

Based on a previous study [14], the number of days alive without antibiotics was expected to be 11 ± 7 days (mean ± standard deviation). A sample size of 450 patients would provide 80% power to detect a 2-day gain in the experimental strategy group, considering a two-sided alpha of 5% and a non-parametric test, as well as a 10% loss to follow-up. We estimated that the targeted sample size could be achieved in 20 centers over 30 months.

Statistical analysis

Baseline characteristics, initial microbiological workup and results were reported using frequencies and percentages for categorical data, and median and interquartile ranges (IQRs) for continuous data.

The primary outcome was compared using a Wilcoxon rank-sum test in the intention-to-treat (ITT) population. For patients with missing outcome data at Day 28 or who died before Day 28, the number of antibiotic-free days was considered zero in both groups.

To account for the over-dispersion of the data, sensitivity analyses were performed using a generalized linear mixed (GLM) model with a binomial-negative distribution and considering center as a random effect, on the ITT and per-protocol (PP) populations. The PP population comprised patients randomized without major protocol violations/deviations (for details, see Supplement 1). *Post-hoc* analyses were performed on the primary outcome with a Desirability of Outcome Ranking (DOOR) methodology (Supplement 1) and in the subpopulation of patients receiving invasive mechanical ventilation at enrollment. The same models were run for the number of antibiotic-free days on Day 7 and Day 14.

Post-hoc analyses also compared the number of patients exposed to each group of antibiotics, the cumulative duration of each group of antibiotics, the proportion of patients with at least one episode of clinically suspected superinfection, the number of patients receiving antibiotics for reasons other than the initial CAP, and the duration of antibiotics according to the microbiological diagnosis from enrollment to Day 28 (for details and definitions, see Supplement 1 and Table S3, Supplement 1). All secondary analyses were compared between groups using median differences (intervention group minus control group) with their 95% confidence intervals (CI) estimated by the Brookmeyer & Crowley method, or proportion differences with their 95% CI estimated by the Exact method.

All analyses were performed with the SAS version 9·4 statistical software (SAS Institute Inc. Cary, NC, USA), except for median differences, for which the CI was estimated with the R Studio version 2022.02.0 + 443 (R: A Language and Environment for Statistical Computing, R Core Team, R Foundation for Statistical Computing, Vienna, Austria, https://www.R-project.org). All tests were two-sided and a *P*-value of less than 0.05 indicated statistical significance. No adjustment for multiple comparisons was made.

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, writing of the report, or decision to submit the paper for publication.

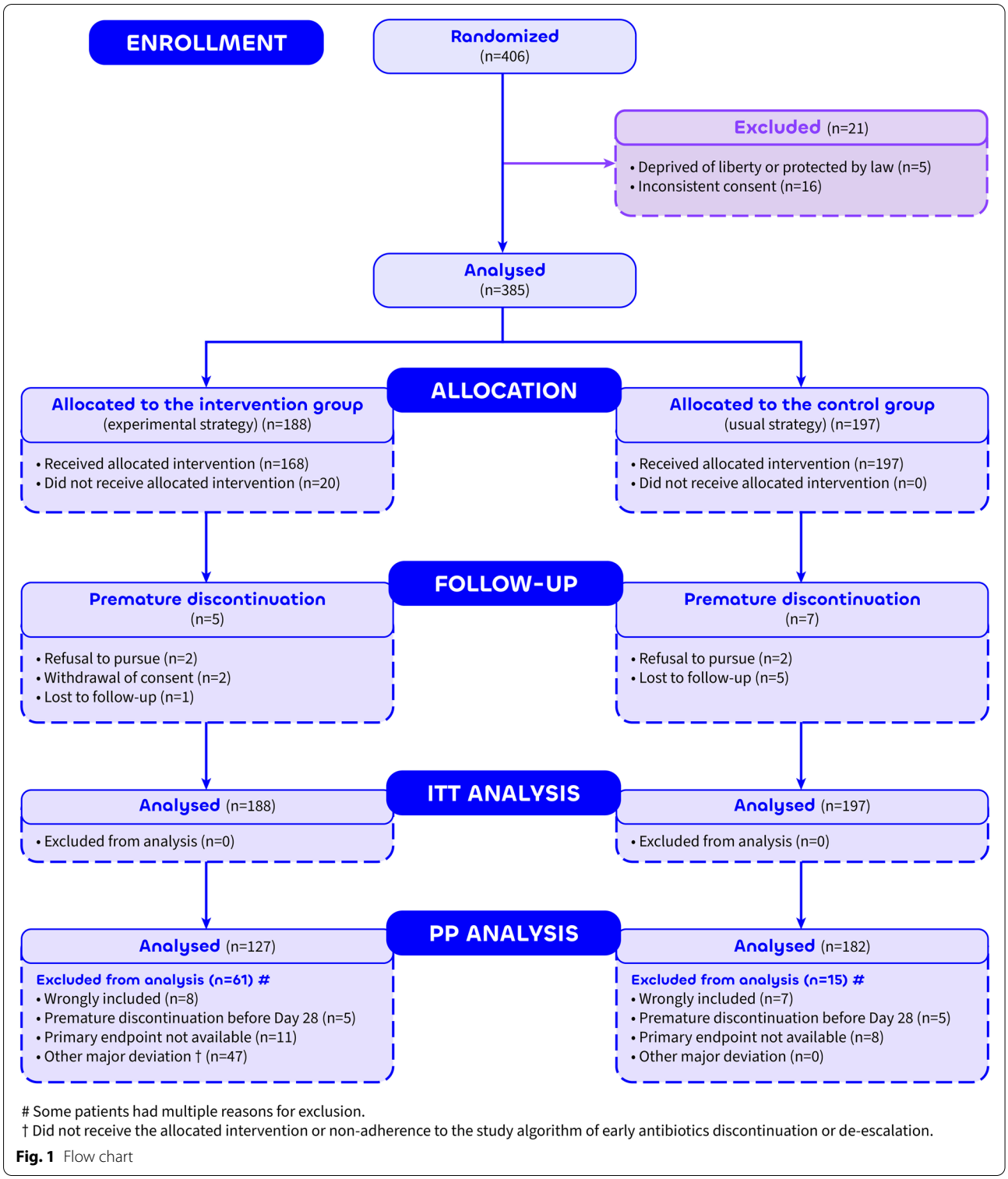
Results

Characteristics of participants

From October 4, 2018 to March 3, 2022, 406 patients were randomized across the 20 participating centers. Recruitment was paused between March 13, 2020 and May 2, 2020 due to the COVID-19 pandemic, and was stopped in March 2022, before reaching the recruitment target, as decided by the sponsor and investigators. Of the 406 randomized patients, 385 were evaluable in the ITT analysis (Fig. 1) (Figure S3, Table S4, Supplement 1). Their baseline characteristics are depicted in Table 1. The median (interquartile range) time from ICU referral to randomization was 9 (3; 16) hours. Prior or ongoing antibiotics were administered to most patients (85%), in a median (interquartile range) time of 1 (0; 1) day before enrollment. No between-group difference was observed.

Initial microbiological work-up and empirical antibiotics

After enrollment, empirical antibiotics were administered to 382 (99%) patients (Figure S4, Table S5, Supplement 1). Narrow-spectrum beta-lactam combined with a macrolide or fluoroquinolone was administered as the initial antibiotic regimen in 255 (66%) patients, with no difference between the two groups (Table S6, Supplement 1). In addition, 90 (23%) patients received steroids as adjunctive therapy (Table S7, Supplement 1). Details of microbiological work-up and documentation are depicted in Table 2. At least 1 lower respiratory tract specimen was obtained for bacteriological exploration in 363 (94%) patients, including 185 (98%) patients in the intervention group and 178 (90%) patients in the control group.



Some patients had multiple reasons for exclusion.
† Did not receive the allocated intervention or non-adherence to the study algorithm of early antibiotics discontinuation or de-escalation.

Fig. 1 Flow chart

At least 1 bacterial microorganism was identified using all conventional tests in 202 (52%) patients, including 103 (55%) patients in the intervention group and 99 (50%) patients in the control group. In the intervention group, no mPCR was performed in three patients, eight patients had an inadequate mPCR performed on a nasopharyngeal swab, and one mPCR result was unavailable. In the remaining 176 patients, mPCR detected no bacteria in 54

Table 1 Baseline characteristics of the patients enrolled in the MULTI-CAP trial, according to the assigned strategy

	Missing, N intervention/control	Experimental strategy (*n* = 188)	Usual strategy (*n* = 197)
Demographics			
Age, years	0/0	65.5 (53.0; 75.0)	67.0 (56.0; 75.0)
Gender, male	0/0	129 (68.6)	140 (71.1)
Anti-influenza vaccine	77/76	15 (13.5)	18 (14.9)
Anti-pneumococcal vaccine	89/89	10 (10.1)	6 (5.6)
COPD			
Asthma	0/1	55 (29.3)	50 (25.5)
Diabetes mellitus	0/1	11 (5.9)	17 (8.7)
Arterial Hypertension	0/1	44 (23.4)	51 (26.0)
Coronary disease	0/1	84 (44.7)	80 (40.8)
Chronic heart failure NYHA III-IV	0/1	27 (14.4)	26 (13.3)
Cerebro-vascular disease	0/1	13 (6.9)	9 (4.6)
Cancer	0/1	21 (11.2)	20 (10.2)
Management before enrollment			
Time from hospitalization to ICU admission (hours)	9/4	4.7 (1.7; 9.7)	4.5 [1.7; 10.3]
Antibiotics received	1/1		
Third generation cephalosporin		133 (71.1)	141 (71.9)
Amoxicillin-clavulanate		29 (15.5)	36 (18.4)
Macrolide		94 (50.3)	120 (61.2)
Fluoroquinolone		10 (5.3)	10 (5.1)
Other medications			
NSAIDs	1/1	16 (8.6)	20 (10.2)
Steroids	1/1	21 (11.3)	11 (5.6)
Severity at enrollment			
Reason for ICU admission	0/0		
Acute respiratory failure		168 (89.4)	178 (90.4)
Sepsis/shock		8 (4.3)	10 (5.1)
Acute renal failure		1 (0.5)	3 (1.5)
Acute neurologic failure		3 (1.6)	0 (0)
Other		8 (4.3)	6 (3.0)
Severity scores SOFA CURB-65 0 to 3 4 to 5	0/0 18/22	6.0 (3.0; 8.0) 122 (71.8) 48 (28.2)	6.0 (3.0; 9.0) 130 (74.3) 45 (25.7)
Vital organ support	0/0		
Conventional oxygen		55 (29.3)	64 (32.5)
High-flow nasal oxygen		35 (18.6)	44 (22.3)
Noninvasive mechanical ventilation		24 (12.8)	22 (11.2)
Invasive mechanical ventilation		64 (34.0)	57 (28.9)
Vasopressors		43 (23.4)	56 (28.9)
Initial laboratory exams			
PaO2 / FiO2, mmHg	15/12	172 (101; 247)	155 (99; 231)
White blood cell count, $\times 10^9$/L	4/8	11.5 (7.6; 17.2)	12.0 (8.3; 16.7)
Platelets count, $\times 10^9$/L	4/10	219 (171; 286)	209 (160; 299)
Procalcitonin, µg/L [§]	3/2	2.1 (0.3; 11.3)	2.5 (0.5; 15.5)
Initial chest x-ray performed Initial chest x-ray findings	0/1 6/7	156 (83.0)	166 (84.7)
Consolidation		84 (56.0)	99 (62.3)
Interstitial opacities		52 (34.7)	64 (40.3)

Table 1 (continued)

	Missing, N intervention/control	Experimental strategy ($n=188$)	Usual strategy ($n=197$)
Bilateral infiltrates		61 (40.7)	67 (42.1)
Pleural effusion		25 (16.7)	23 (14.5)

Data are *n* (%) or median (interquartile range)

§ Value of procalcitonin refers to the first sample obtained in the 24 h following randomization

COPD chronic obstructive pulmonary disease; *CURB-65* confusion, uremia, respiratory rate, blood pressure, age ≥ 65 years; *ICU* intensive care unit; NSAIDs, non-steroidal anti-inflammatory drugs; *NYHA* New York Heart Association; *SOFA* sequential organ failure assessment; PaO_2 / FiO_2, ratio of partial pressure of arterial oxygen (PaO_2) to the fraction of inspired oxygen (FIO_2)

(31%) patients, one bacterium in 68 (39%) patients and 2 or more bacteria in 54 (31%) patients (Table S8, Supplement 1). Overall, at least 1 bacterial microorganism was identified using all microbiological tests in 147 (78%) patients in the intervention group, and 99 (50%) patients in the control group (difference, 27.9 (95% CI, 18.4 to 37.0)). Predominant bacteria were *Streptococcus pneumoniae* and *Hemophilus influenzae,* identified in 95 (25%) and 73 (19%) patients, respectively. Interestingly, *Staphylococcus aureus* ($n=52$), *Legionella pneumophila* ($n=16$) and/or *Pseudomonas aeruginosa* ($n=18$) were identified in 58 (31%) patients in the intervention group, as compared with 22 (11%) patients in the control group (difference, 19.7 (95% CI, 11.4 to 27.8)). At least 1e respiratory virus was detected in 42 (22%) patients in the intervention group, and 29 (15%) patients in the control group (difference, 7.6 (95% CI, -0.2 to 15.6)).

Observance of the experimental strategy

The implementation of the experimental strategy, including adherence to the algorithm for early antibiotics discontinuation or de-escalation, was quite high (141/188; 75%) (for definition and details, see Supplement 1).

Primary outcome

On Day 28 after enrollment, the median number of days alive without antibiotics was 19.0 (0.0; 24.0) days in the intervention group, and 19.0 (7.0; 22.0) days in the control group (difference, 0.0 (95% CI, -4.0 to 4.0); p = 0.70) (Figs. 2 and 3, Table 3). Similar results were obtained in the PP population, confirmatory analysis (Table S9, Supplement 1) and DOOR analysis (Supplement 1).

Secondary outcomes

On Day 28 and Day 90, there was no difference between the two groups for the secondary outcomes, except for the cumulative duration of antibiotics which was 3 days shorter in the intervention group (95% CI, − 5.1 to − 0.9) compared with the control group in the ITT analysis (Table 3). This result was consistent with the sensitivity analysis restricted to the PP population and the subpopulation of patients receiving invasive mechanical ventilation at enrollment (Table S10, Supplement 1). Medico-economic evaluation is detailed in Supplement 1 and Figure S5 (Supplement 1).

Exploratory and post-hoc analyses

The number of antibiotic-free days was 1 day higher (risk-ratio 1.40 (95% CI, 1.03 to 1.91)) on Day 7 (Table S11, Figure S6, Supplement 1) and the median time from enrollment to first interruption of all antibiotics was 1 day lower (95% CI, − 1.7 to − 0.3) in the intervention group compared to the control group (Table S12, Supplement 1). The cumulative duration of narrow-spectrum beta-lactams on Day 28 was 1 day lower (95% CI, − 1.7 to − 0.3) in the intervention group compared to the control group (Table S13, Supplement 1). Post-hoc analyses based on the bacterial documentation, procalcitonin levels from enrollment to Day 7, and antibiotic-free days after Day 10 are presented in Supplement 1 (Figure S7, Table S14, Table S15, Supplement 1).

The experimental strategy did not alter the proportion of patients with at least one episode of superinfection from enrollment to Day 28 (Table S16, Supplement 1) or the time to the first antibiotic regimen for another reason than CAP (Figure S8, Supplement 1). Furthermore, there was no between-group difference in the proportion of patients exposed to broad-spectrum antibiotics (Table S17, Supplement 1) or in the proportion of patients requiring ventilatory support from enrollment to Day 28 (Table S18, Supplement 1).

Safety issues

Serious adverse events did not differ between groups (Tables S19, S20 and S21, Supplement 1).

Discussion

This open-label randomized controlled trial investigated the efficacy and safety of a strategy combining broad-spectrum respiratory mPCR, procalcitonin measurements, and an algorithm for early antibiotic discontinuation or de-escalation in ICU patients with CAP,

Table 2 Microbiological work-up and results in patients enrolled in the Multi-CAP trial, according to the assigned strategy

	Experimental strategy (*n* = 188)	Usual strategy (*n* = 197)
At enrollment *		
Conventional tests, positive/performed and analyzed (n/N) (%)		
Blood cultures	28/169 (16.6)	26/183 (14.2)
Legionella pneumophila urinary antigen test	9/165 (5.5)	4/182 (2.2)
Streptococcus pneumoniae urinary antigen test	29/132 (22.0)	22/154 (14.3)
Sputum	34/81 (42.0)	25/86 (29.1)
Tracheal aspirate or protected distal sample	40/63 (63.5)	35/70 (50.0)
Bronchoalveolar lavage	14/33 (42.4)	15/35 (42.9)
Atypical bacteria simplex PCR	3/17 (17.6)	3/15 (20)
Viral PCR [§]	13/103 (12.6)	29/140 (20.7)
Bacteria identified (all conventional tests), *n* (%)		
Streptococcus pneumoniae	43 (22.9)	37 (18.8)
Staphylococcus aureus	19 (10.1)	12 (6.1)
Haemophilus influenzae	18 (9.6)	15 (7.6)
Enterobacterales	29 (15.4)	25 (12.7)
Pseudomonas aeruginosa	8 (4.3)	5 (2.5)
Legionella pneumophila	9 (4.8)	5 (2.5)
Mycoplasma pneumoniae	3 (1.6)	2 (1.0)
Other bacteria	33 (17.6)	30 (15.2)
Virus identified (all conventional tests), *n* (%)		
Influenza Virus	8 (4.3)	22 (11.2)
Non-Influenza virus	5 (2.7)	7 (3.6)
mPCR after enrollment		
mPCR positive for at least one bacteria/performed and analyzed, *n* (%) [#]		
Sputum	67/89 (75.3)	
Tracheal aspirate	44/67 (65.6)	
Broncho-alveolar lavage	10/21 (47.6)	
Bacteria detected, n (%)		
At least one bacterial micro-organism	122 (64.9)	
Streptococcus pneumoniae	49 (26.1)	
Staphylococcus aureus	36 (19.1)	
Haemophilus influenzae	57 (30.3)	
Enterobacterales	25 (13.3)	
Pseudomonas aeruginosa	10 (5.3)	
Legionella pneumophila	7 (3.7)	
Mycoplasma pneumoniae	3 (1.6)	
Moraxella catarrhalis	14 (7.4)	
Other bacteria*	8 (4.3)	
Resistance genes, *n* (%)		
CTX-M	0 (0)	
IMP, KPC, NDM, OXA-48-like, VIM	0 (0)	
Mec-A/C + MREJ	6 (3.2)	
Respiratory virus detected, *n* (%)		
At least one virus	41 (21.8)	
Human Rhinovirus—Enterovirus	17 (9.0)	
Influenza virus	10 (5.3)	
Respiratory syncytial virus	3 (1.6)	
Human metapneumovirus	4 (2.1)	

Table 2 (continued)

	Experimental strategy ($n = 188$)	Usual strategy ($n = 197$)
Coronavirus (non-SARS-CoV-2)	4 (2.1)	
Parainfluenza virus	5 (2.7)	

Data are n (%) or n/N (%)

[§] Viral PCR were either simplex (Influenza virus) or multiplex, and were performed on either nasopharyngeal swabs or distal samples (sputum, tracheal aspirate, broncho-alveolar lavage)

[#] Three patients had no mPCR performed and eight patients had a mPCR inadequately performed on a nasopharyngeal swab

mPCR multiplex polymerase chain reaction, *mPCR* SARS-CoV-2; severe acute respiratory syndrome coronavirus 2. Resistance genes are CTX-M (cefotaxime-Munich), *IMP* imipenemase, KPC (*Klebsiella pneumoniae* carbapenemase), *NDM* New Delhi metallo-β-lactamase), *OXA-48* oxacillinase 48), VIM (Verona integron-encoded metallo-β-lactamase), MEC (methicillin resistance) and MREJ (Mec-associated recombinational junction)

compared to standard care. The combined approach did not reduce the overall antibiotic exposure or improve the clinical success on Day 28. However, it resulted in a 3-day shorter antibiotic cumulative duration on Day 28, and a one-day shorter cumulative duration of narrow-spectrum beta-lactams on Day 28.

Previous series of CAP in the ICU usually reported bacterial identification rates around 50% with standard culture-dependent methods [17–19]. In our study, we aimed to evaluate the benefits of a respiratory mPCR. Obtaining a lower respiratory tract specimen was crucial so that more than four patients out of five were sampled. Consequently, despite the fact that antibiotics were administered frequently prior to ICU admission, the bacterial detection rate reached 52% with conventional testing. When adding the mPCR to the conventional

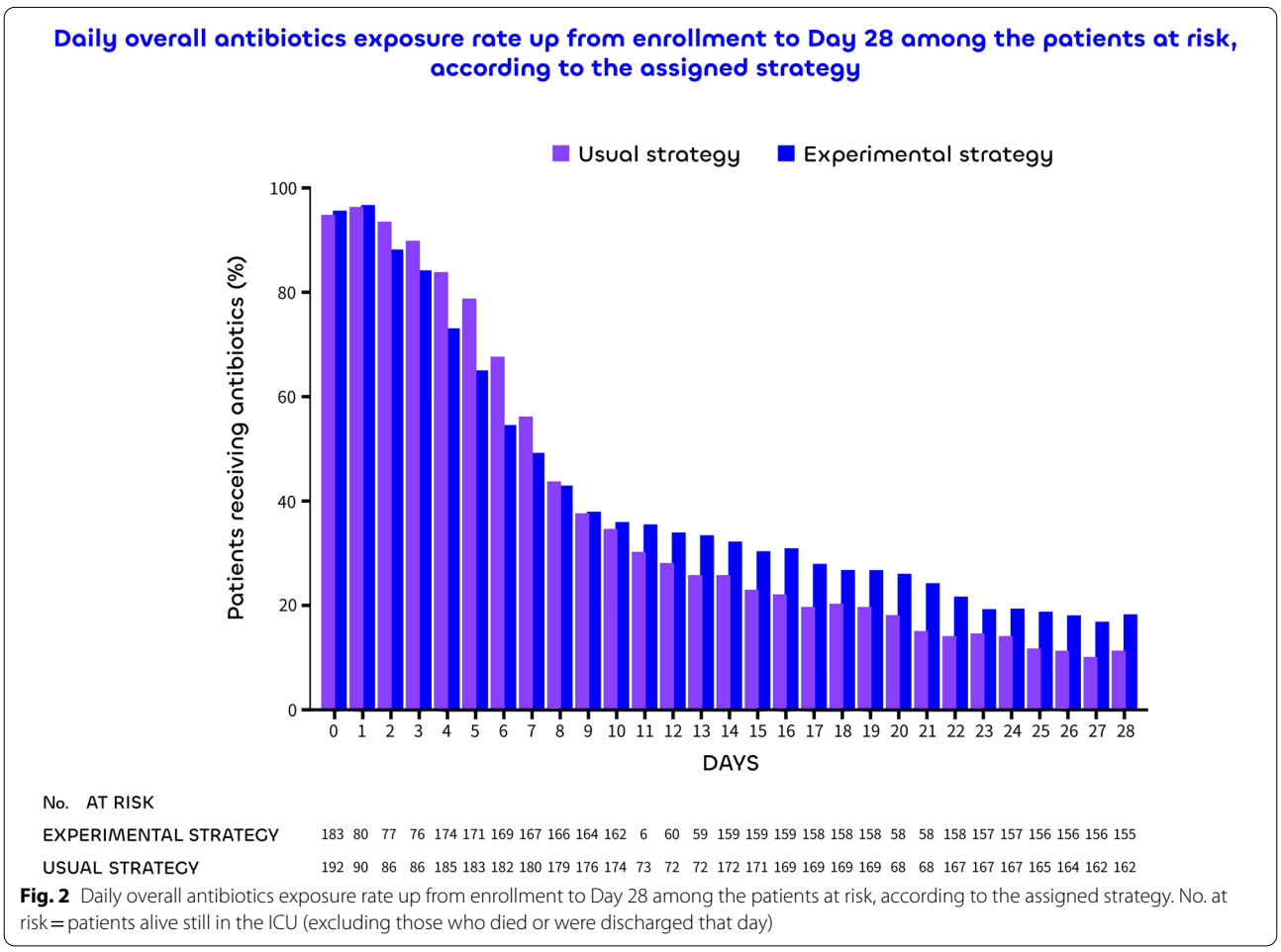


Fig. 2 Daily overall antibiotics exposure rate up from enrollment to Day 28 among the patients at risk, according to the assigned strategy. No. at risk = patients alive still in the ICU (excluding those who died or were discharged that day)

Table 3 Primary and secondary outcomes of patients enrolled in the Multi-CAP trial, according to the assigned strategy

Endpoints	Experimental strategy ($n=188$)	Usual strategy ($n=197$)	Difference, (95% CI)	*P*
Primary outcome at Day 28				
Number of antibiotic-free days in the ITT population, days	19.0 (0.0; 24.0), $n=188$	19.0 (7.0; 22.0), $n=197$	0.0 (− 4.0; 4.0)	0.70
Number of antibiotic-free days in the PP population, days	20.0 (4.0; 24.0), $n=127$	19.5 (9.0; 22.0), $n=182$	0.5 (-3.8; 4.8)	0.71
Secondary outcomes at Day 28				
Mortality rate, *n* (%)	28/183 (15.3)	30/192 (15.6)	− 0.3 (− 7.8; 7.2)	
Death or persistent organ failure (SOFA > 0) at Day 28, *n* (%)	39/146 (26.7)	38/153 (24.8)	1.9 (− 8.2; 12.0)	
Antibiotic exposure [#], days	7.0 (4.0; 16.0), $n=178$	8.0 (6.0; 13.5), $n=188$	− 1.0 (− 2.6; 0.6)	
Cumulative duration of antibiotics [∞], days	10.0 (6.0; 22.0), $n=178$	13.0 (8.0; 20.0), $n=188$	− 3.0 (− 5.1; − 0.9)	
Colonization/infection with multidrug-resistant bacteria or *Clostridium difficile*, *n* (%)	6/147 (4.1)	10/149 (6.7)	− 2.6 (− 8.4; 3.0)	
Length of ICU stay, days	6.0 (4.0; 11.0), $n=188$	7.0 (3.0; 12.5), $n=196$	− 1.0 (− 2.4; 0.4)	
Length of hospital stay, days	16.0 (9.0; 29.0), $n=175$	16.5 (10.0; 26.5), $n=176$	− 0.5 (− 4.4; 3.4)	
Secondary outcomes at Day 90‡				
Overall mortality rate	32/183 (17.5)	40/190 (21.1)	− 3.6 (− 11.7; 4.6)	
Total cost of care [§], €	25,134 ± 21,012, $n=188$	24,647 ± 18,235, $n=197$	232 (− 3,702; 4,166)	
Health status (VAS 0–100) [¶]	66.0 ± 24.3, $n=41$	64.8 ± 21.2, $n=56$	1.1 (− 8.1; 10.4)	
QALYs (EQ5D3L score)	0.03 ± 0.06, $n=45$	0.02 ± 0.06, $n=57$	0.003 (− 0.007; 0.015)	

Data are *n* (%), mean ± sd, or median (interquartile range)

[#] Antibiotic exposure was measured as the number of days with at least one antibiotic (regardless the route and dose regimen) from enrollment to Day 28

[∞] The cumulative duration of antibiotics, as a measure of the total antibiotic consumption, was calculated as the sum of the duration of each antibiotic from enrollment to Day 28

[§] Restricted to hospital care (index hospitalization and in-trial follow-up period)

[¶] Health status (VAS 0–100) is a component of the EQ5D3L

EQ5D3L EuroQuol 5 dimensions questionnaire; *ICU* intensive care unit; *ITT* intention-to-treat; *PP* per-protocol; *QALYs* quality-adjusted life years; *SOFA* Sequential Organ Failure Assessment; *VAS* visual analogic scale

microbiological work-up, this rate increased to an unusually high rate of 78%. This finding underscores the value of molecular testing in etiological diagnosis, consistent with prior studies of severe CAP in the ICU [7, 20] or SARS-CoV-2-associated pneumonia [21]. However, the use of mPCR testing presents challenges for clinicians, particularly in interpreting results, which are often multi-positive for bacterial microorganisms—especially when performed on sputum, as previously reported [22, 23]. Furthermore, except for atypical bacteria, results are reported in copies/mL. However, the thresholds between colonization and infection remain unclear, and the correlation with CFU/mL is imperfect [24]. Due to these complexities, some researchers have chosen to pair mPCR use with antimicrobial stewardship to assist clinicians in interpreting the results and to facilitate the transition from empirical antibiotics to pathogen-directed therapies [7, 25, 26]. Here, we chose not to add another layer to the decision-making process, opting instead to allow clinicians to manage the mPCR results directly, as they usually do with whole microbiological investigations, taking into account to the clinical context. Therefore, we developed a novel algorithm combining the results of both mPCR testing and serial procalcitonin serum measurements. A similar approach was tested in a randomized controlled trial involving adult inpatients with non-pneumonic lower respiratory tract infections [27]. However, this trial, which used a viral-panel mPCR, did not show a significant reduction in antibiotic exposure. Our algorithm was specifically designed to identify patients in whom a bacterial infection can be confidently ruled out, based on a comprehensive microbiological work-up that includes mPCR testing and a procalcitonin level below 1 ng/mL. Moreover, the algorithm promoted de-escalation of antibiotics when bacteria was detected, encouraging a shift from empirical (dual) therapy to pathogen-directed therapy, in accordance with written antibiotic guidelines. Clinicians quickly adopted and understood this approach, as adherence to the algorithm-based strategy was quite high, did not vary over time, and did not differ between major and minor contributing centers. Interestingly, adherence to the algorithm-based strategy was lower when the mPCR did not detect bacteria. This finding was consistent with previous studies on non-pneumonia lower respiratory tract infections [27] and severe SARS-CoV-2 pneumonia [21], which showed

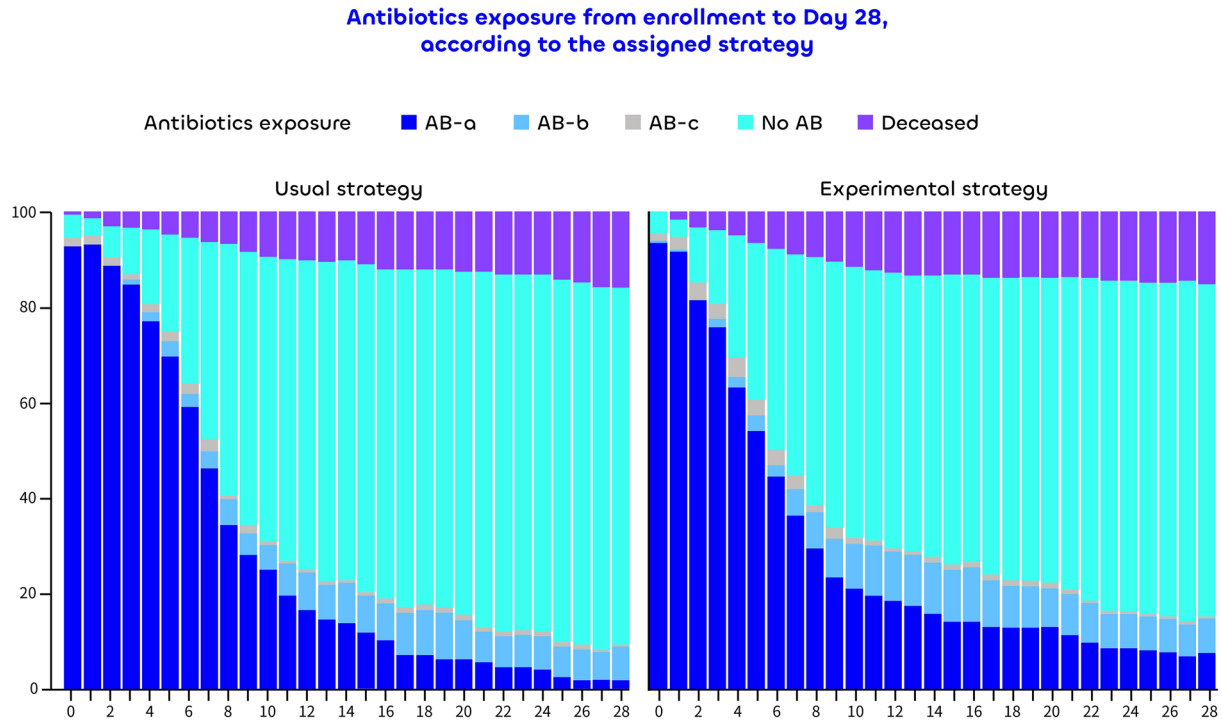


Fig. 3 Antibiotics exposure from enrollment to Day 28, according to the assigned strategy. *AB* antibiotics; *sCAP* severe community-acquired pneumonia. *AB-a* antibiotics for the sCAP, ii) AB-b = antibiotics for a superinfection, iii) AB-c = antibiotics for the sCAP and a superinfection

that PCR-based strategies can yield better adherence in patients with microbiological documentation. Finally, we found that there was potential to reduce antibiotic exposure if clinicians had systematically applied the algorithm and followed its advice. However, they only did so for six out of seven patients. Similar findings have been reported in previous studies assessing procalcitonin-based approaches to reduce antibiotic duration in patients with lower respiratory tract infection [28] or ventilator-associated pneumonia [29]. Several reasons have been raised. Clinicians may believe in bacterial infection despite negative biomarkers and/or microbiological results. Other reasons may include inexperience in applying algorithms or protocol guidance, and a lack of culture surrounding the active de-escalation or early interruption of antibiotics. These elements all highlight residual barriers between the implementation of rapid diagnostic testing and improved antimicrobial stewardship.

We chose a primary outcome that considered both prognosis and antibiotic exposure. Indeed, previous studies addressing the impact of mPCR testing in hospitalized patients with lower respiratory tract infection primarily focused on antibiotic management alone. Typically, these studies measured the time to first antibiotic modification [26], the time to appropriate therapy [25] or the time to pathogen-directed therapy [7, 9]. In the only trial that used such a dual primary outcome, a management strategy combining point-of-care mPCR testing and an antimicrobial prescribing algorithm resulted in an unexpected lower frequency of clinical cure on Day 14 in adult or infant ICU patients with hospital-acquired or ventilator-associated pneumonia [30]. However, adherence to the algorithm in the intervention group was only 31% and the cure rate was higher in patients for whom the algorithm was followed. In our study, there were no safety concern regarding mortality on Day 28 and Day 90. In terms of antibiotic exposure, we observed a 1-day reduction in antibiotic exposure over the first 7 days, in line with findings from our previous work involving ICU adults with SARS-CoV-2-related pneumonia [21]. The duration of macrolide treatment was remarkably brief in both groups, indicating that early de-escalation by discontinuing the anti-atypical pathogens was widely implemented in both groups. Therefore, the positive signal on Day 7 was more likely attributable to the earlier discontinuation of narrow-spectrum beta-lactams in the intervention group, since the total exposure was reduced by 1 day on Day 28. However, this difference in antibiotic exposure was no longer observed on Day 28. This finding was not attributed to a higher incidence of superinfections in the intervention group. An increased detection of bacteria that might have indicated

a prolonged antibiotics course is a possible explanation. Indeed, the number of patients documented with *Staphylococcus aureus*, *Legionella pneumophila,* and/or *Pseudomonas aeruginosa* was almost three times higher in the intervention group. As expected, we observed higher antibiotic exposure in these patients, regardless of the assigned strategy. In summary, the mPCR-based approach allowed for antibiotic sparing in some patients, while in others it prompted microbiologically indicated prolonged antibiotic courses. This ultimately resulted in median antibiotic exposure at Day 28 comparable to that observed in the control group.

Our study had several limitations. First, the algorithm did not account for bacterial quantification (copies/mL) in mPCR testing, as incorporating this would have made the algorithm overly complex. The correlation between the number of copies and bacterial inoculum is still unclear and did not allow the enrollment of mPCR quantification in the therapeutic algorithm. Secondly, using procalcitonin to guide antibiotics discontinuation after Day 3 in both groups, combined with a recommended maximum duration of antibiotics of 7 days may have contributed to minimize the difference between groups. This likely contributed to the relatively short median time to the first discontinuation of all antibiotics in the control group compared with the previous series of CAP [14, 31]. Nevertheless, this approach aligns with recent guidelines [6], suggesting the use of procalcitonin levels to guide antibiotic discontinuation in patients with CAP [4]. Another interesting approach would have been to use the combination of C-reactive protein and procalcitonin, which may be better than procalcitonin alone [32]. Thirdly, our written guidelines for antibiotic therapy recommended a longer duration for some difficult-to-treat pathogens such as *Staphylococcus aureus* and *Pseudomonas aeruginosa*. This can be criticized because such recommendation is not universal and is supported by low-quality evidence [6]. Such an approach may have contributed to prolonged antibiotic exposure in a significant number of patients in the intervention group. Fourthly, the study was conducted in France, a country with a low incidence of methicillin-resistant *Staphylococcus aureus* (MRSA). It is plausible that an mPCR-based approach could provide greater benefits in terms of broad-spectrum antibiotic sparing in countries with a higher incidence of MRSA. Likewise, our study involved immunocompetent patients with a low risk of resistant bacteria like *Pseudomonas aeruginosa*. We cannot exclude a more important reduction of broad-spectrum antibiotics with mPCR in other populations with a higher risk of infection with multiresistant bacteria [25]. Finally, the number of antibiotic-free days observed was higher than expected when calculating the sample size for this study, possibly leading to insufficient statistical power.

Overall, the experimental strategy combining a broad-spectrum respiratory mPCR and procalcitonin to guide antibiotic de-escalation or discontinuation in ICU patients with CAP failed to reduce antibiotic exposure and improve outcomes on Day 28, compared to the usual strategy. However, despite negative results on antibiotic-free days on Day 28, the intervention may be associated with early antibiotic savings and a reduction in cumulative antibiotic duration on Day 28.

Supplementary Information

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Guillaume Voiriot, Isabelle Durand-Zaleski, Laurence Berard, Tabassome Simon, Muriel Fartoukh and Jean-François Tismit conceived and designed the study. Guillaume Voiriot, Laurent Argaud, Yves Cohen, Louis Chauvelot, Bertrand Souweine, Kada Klouche, Jean Reignier, Carole Schwebel, Anahita Rouzé, Armand Mekontso Dessap, Julien Bohé, Bruno Megarbane, Julien Carvelli, Jean-Christophe Navellou, Sébastien Gibot, Éric Maury, Jean Dellamonica, Pierre-François Dequin, Julien Dessajan, Muriel Fartoukh and Jean-François Tismit recruited patients. Laurence Armand-Lefèvre, Francois Vandenesch and Charlotte Verdet performed microbiological investigations. Guillaume Voiriot, Sophie Tuffet, Isabelle Durand-Zaleski, Alexandra Rousseau, Simon Tabassome, Muriel Fartoukh and Jean-François Tismit analyzed and interpreted the data. Guillaume Voiriot, Sophie Tuffet, Alexandra Rousseau, Muriel Fartoukh and Jean-François Tismit drafted the article. All authors participated in the critical revision of the manuscript and provided final approval to submit this version of the manuscript.

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Data availability
Consultation by the editorial board or interested researchers may be considered, subject to prior determination of the terms and conditions of such consultation and in respect for compliance with the applicable regulations.

Declarations

Conflicts of interest
Laurence Armand-Lefèvre received consulting fees from Shionogi and Viatris, and honoraria for lectures from Advanz, Shionogi and Pfizer. In addition, she performed unpaid lectures for BioMérieux and Qiagen. Julien Dessajan received honoraria for lectures from BioMérieux. Isabelle Durand-Zaleski received consulting fees from Bristol Myers Squibb, MSD, Sanofi, Pfizer and Roche. Muriel Fartoukh reported research grants from BioMérieux and personal fees from BioMérieux, Pfizer, SOS Oxygene and Fisher and Paykel. Tabassome Simon reported support for research from BioMérieux, research grants from Astra Zeneca, Bayer, Boehringer, Daiichi Sankyo, Ili-Lilly, GSK, Novartis and Sanofi, and honoraria for lectures from Servier and Novartis. In addition, she reported participation in Data Safety Monitoring Board for Ablative Solutions, Air Liquide, Astra Zeneca, Sanofi, Novartis and 4Living Biotech. Jean-Francois Timsit Simon reported support for research from BioMérieux, and consulting fees from Advanz, Menarini, Pfizer and Merck. In addition, he reported honoraria for lectures from BioMérieux, MundiPharma and Merck. Guillaume Voiriot reported research grants from BioMérieux and SOS Oxygene, and support for attending meetings from Oxyvie. Francois Vandenesch reported been founder and shareholder of Weezion.

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